

Multi-person Physics-based Pose Estimation for Combat Sports

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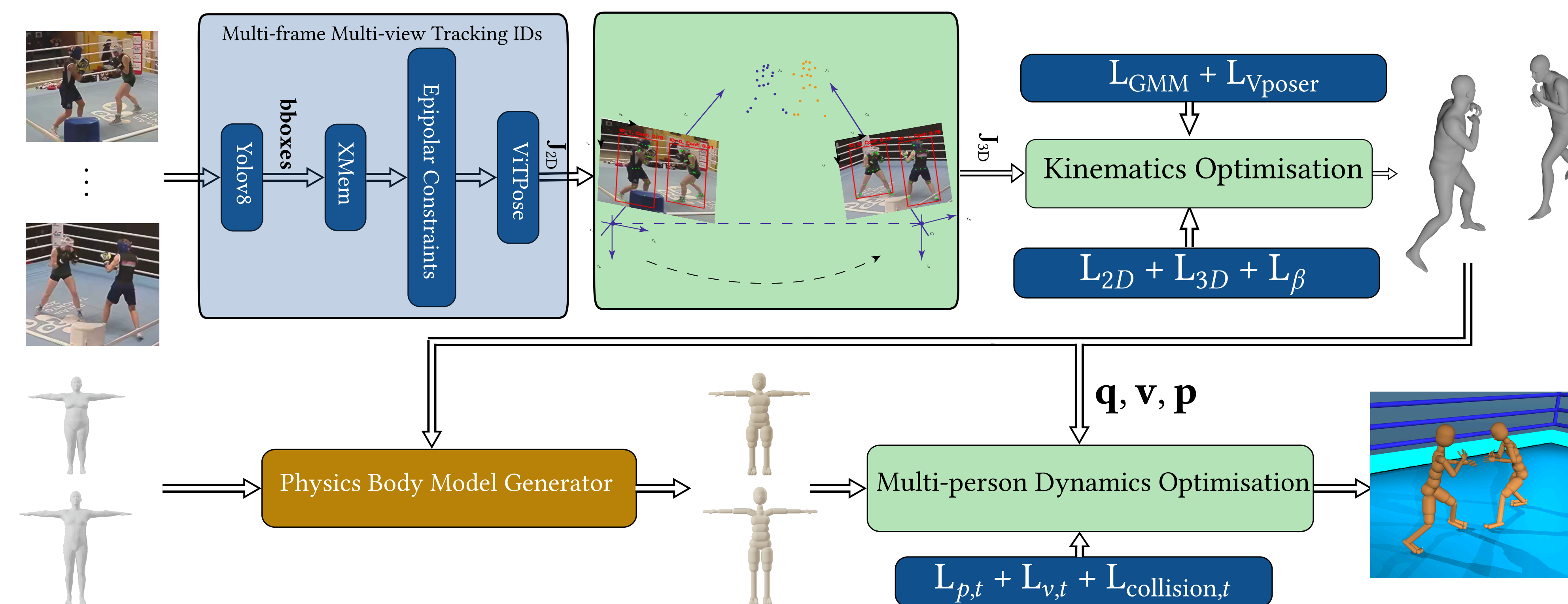
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Introduction

Motion capture in sports scenes like boxing presents challenges due to occlusion, crowding, and fast movements. We use a multi-view setup that requires visibility from only two cameras. Our pipeline produces realistic motions through multi-person physics optimisation while accounting for inter-character collisions.

Pipeline



The pipeline begins with bounding boxes (**bboxes**) and robust tracking IDs. 2D poses (J_{2D}) from ViTPose [xu2022vitpose] are triangulated to 3D (J_{3D}). Kinematics optimisation fits SMPL (θ, β), then an iLQR physics stage [howell2022] corrects artefacts using state ($\mathbf{q}, \mathbf{v}, \mathbf{p}$).

Multi-frame Multi-view Tracking IDs

- **Tracking by segmentation:** XMem memory for stable IDs [cheng2022xmem]
- **Epipolar ID matching:** Centroid & epipolar constraints [wohler20093d]
- **Top-down 2D poses:** ViTPose inside each box [xu2022vitpose]

Weighted Triangulation and Filtering

Weighted triangulation solves

$$\mathbf{A} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0, \quad \mathbf{A} = \begin{bmatrix} \mu_1(P_{11} - u_1P_{31}) \\ \mu_1(P_{21} - v_1P_{31}) \\ \vdots \\ \mu_N(P_{1N} - u_NP_{31N}) \\ \mu_N(P_{2N} - v_NP_{32N}) \end{bmatrix}.$$

SVD yields (x, y, z) . Cubic-spline smoothing removes outliers; an EKF fills missing frames when triangulation fails.

Kinematics Optimisation

Optimise SMPL θ by

$$\min_{\theta} \sum_t (w_1 L_{2D} + w_2 L_{3D} + w_3 L_{\text{reg}} + w_4 L_{\text{smooth}} + w_5 L_{\text{GMM}} + w_6 L_{\text{Vposer}}).$$

Physics Body Model Generator

For K humanoids the state is $\mathbf{x}_t = (\mathbf{q}_t, \mathbf{v}_t) \in \mathbb{R}^{125K}$, with $\mathbf{p}_t \in \mathbb{R}^{21K}$ SMPL joints. We export to MuJoCo XML [todorov2012mujoco].

Multi-person Dynamics Optimization

The iLQR algorithm [howell2022] optimizes a control trajectory $\mathbf{u}_{0:T}$ by minimizing the cost function. The reference positions. $\mathbf{p}_t$ and velocities $\mathbf{v}_t$ are extracted from the kinematics optimization at each frame t , and the trajectory optimization is "warm-started" using control parameters $\mathbf{u}_{t-1}$ from the previous frame. The control is updated iteratively as $\Delta \mathbf{u}_t = \mathbf{K}_t \Delta \mathbf{x}_t + \alpha \mathbf{k}_t$, with $\mathbf{K}_t$ and $\mathbf{k}_t$ refined until convergence.

$$\text{Dynamics Optimization Formulation: } \min_{\mathbf{u}_{0:T}} \sum_{t \in [0, T]} w_1 L_{\text{reg},t} + w_2 L_{\mathbf{p},t} + w_3 L_{\mathbf{v},t} + w_4 L_{\text{collision},t}$$

- **Position Loss:** Minimizes the difference between predicted and actual 3D joint positions in each short horizon.

$$L_{\mathbf{p},t} = \sum_{i \in J_{3D}} \|\mathbf{p}_{i,t} - \mathbf{J}_{3D,i,t}\|^2$$
- **Velocity Loss:** Minimizes the difference between predicted and actual 3D joint velocities in each short horizon.

$$L_{\mathbf{v},t} = \sum_{i \in J_{3D}} c_i \|\mathbf{v}_{i,t} - \mathbf{v}_{3D,i,t}\|^2$$
- **Collision Loss:** The collision term penalizes the penetration between the geoms of different humanoids.

$$L_{\text{collision},t} = \sum_{i \in 1 \dots M} \sum_{j \in 1 \dots M} \max(0, \epsilon - \text{dist}(G_i, G_j))$$
- **Regularization Loss:** Prevent sudden changes in control input. $L_{\text{reg},t} = \|\mathbf{u}_t\|^2$

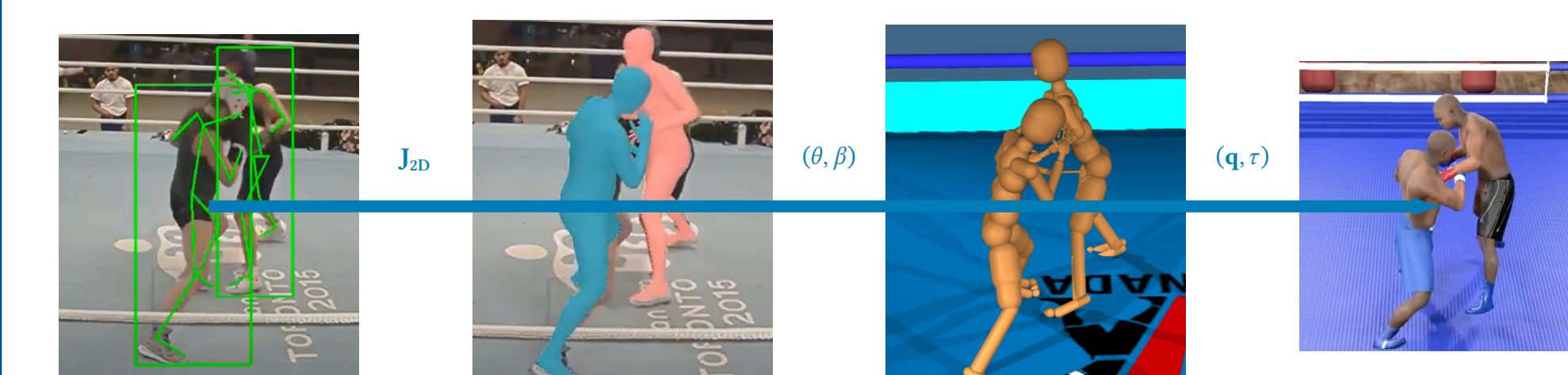
Discussion and Future Works

We captured a 10-hour boxing dataset to enable real-time inference and richer contact modeling.

Video



Visualisation



References